MIL-H-51147A(MU)
7 November 1966
SUPERSEDING
MIL-H-51147(MU)
19 July 1963

MILITARY SPECIFICATION

HERBICIDE; n-BUTYL 2,4-DICHLOROPHENOXYACETATE

1. SCOPE

1.1 This specification covers one type of herbicide consisting of normal-butyl 2,4-dichlorophenoxyacetate (2,4-D), molecular weight 277.15.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-729 - Drums: Metal, 55-Gallon (For Shipment of Non-Corrosive Material).

FSC 6840

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF OFFICIAL AGRICULTURAL CHEMISTS

Official Methods of Analysis of the Association of Official Agricultural Chemists (AOAC), Ninth Edition, 1960.

(Application for copies should be addressed to the Association of Official Agricultural Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, D. C. 20004.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Standards

D664 - Method of Test for Neutralization Number by Potentiometric Titration.

D1533 - Method of Test for Water in Insulating Liquids (Karl Fischer Method).

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race St., Philadelphia 8, Pennsylvania 19103.)

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UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (USP), Sixteenth Revision, 1960.

(Copies of the USP may be obtained from the Mack Publishing Company, Easton, Pennsylvania 18042.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

CODE OF FEDERAL REGULATIONS

7CFR 362 - Agriculture, Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act.

(Copies of the above part of the Code of Federal Regulations may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20025.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Composition. The herbicide shall be composed of no less than 95 percent normal-butyl 2,4-dichlorophenoxyacetate.

3.1.2 Appearance. The herbicide shall be a clear, brown, liquid.

3.2 Total acid equivalent. The total acid equivalent of the herbicide shall be no less than 76 nor more than 80 percent when tested as specified in 4.6.1.

3.3 Free acid. The free acid content of the herbicide shall not be greater than 5.0 percent when tested as specified in 4.6.2.

3.4 Undissolved matter. The herbicide shall be completely soluble in reagent-grade benzene, and shall show no evidences of undissolved matter when tested as specified in 4.6.3.

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3.5 Moisture content. The moisture content of the herbicide shall not be greater than 0.2 percent when tested as specified in 4.6.4.

3.6 Specific gravity. The specific gravity of the herbicide shall be no less than 1.238 nor more than 1.253 at 20°/20°C when tested as specified in 4.6.5.

3.7 Preproduction sample. When specified in the contract or purchase order, a preproduction sample of the herbicide shall be produced in accordance with this specification for examination and tests. (See 6.1.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedures which will insure equal or better assurance of quality by submitting a written proposal, with justification, and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification shall govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lotting. A lot shall consist of the herbicide produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.4 Sampling for test. Sampling shall be conducted in accordance with MIL-STD-105, level S-1.

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4.5 Inspection provision. A sample shall be removed from each sample container and placed in a clean, dry container labeled to identify the lot and container from which it was taken. Each sample shall be tested in accordance with 4.6 and MIL-STD-105, level S-1, with an AQL of 2.5 percent defective.

4.6 Tests. Tests shall be conducted as follows.

4.6.1 Total acid equivalent. Determine the total 2,4-D acid in the herbicide in accordance with the procedures specified in paragraphs 4.130, 4.131, 4.132, and 4.161(a) or 4.161(c) of AOAC. Calculate the percent total acid equivalent from the weight of the 2,4-D acid, calculated from the determined chlorine value, by multiplying by 100 and dividing by the weight of the sample.

4.6.2 Free acid. Determine the percent free acid using the apparatus and procedure for strong acids specified in ASTM Designation D664 using USP ethyl alcohol as a solvent. Titrate to pH 4.5. Calculate the percent free acid using the formula:

$$A = \frac{22.10\ VN}{W}$$

where A = percent 2,4-D acid (free acid)

N = normality of 0.1 N alcoholic potassium hydroxide

V = milliliters of 0.1 N alcoholic potassium hydroxide

W = weight in grams of sample.

4.6.3 Undissolved matter. Dissolve 10 parts of the sample by volume in 40 parts of reagent-grade benzene (C_6H_6). Thoroughly agitate the mixture and visually examine for undissolved matter.

4.6.4 Moisture content. Determine the moisture content of the sample in accordance with ASTM Designation D1533.

4.6.5 Specific gravity. Determine the specific gravity of the sample by means of a suitable hydrometer, or by any other standard procedure which is accurate to four significant figures.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. The herbicide shall be packed in a 55-gallon drum conforming to type II of PPP-D-729.

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5.1.2 Level C. Fifty-five gallons of herbicide shall be packed in a manner to insure carrier acceptance and safe delivery at first destination. Containers shall be in accordance with Uniform Freight Classification Ratings, Rules, and Regulations or regulations of other carriers applicable to the mode of transportation. Drums shall not affect nor be affected by the material contained.

5.2 Marking. Each shipping container shall be marked "FOR CONTROLLED MILITARY USE ONLY. NOT FOR RESALE". in a manner conforming to MIL-STD-129. In addition to any other special marking required by the contract or order, each shipping container shall also be marked in accordance with MIL-STD-129 and in compliance with The Federal Insecticide, Fungicide, and Rodenticide Act and other applicable existing Federal laws.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Level of packing required.

(c) Preproduction requirements (when required).

1. Quantity required.

2. Time allowed for submission of samples for Government test and evaluation after award of contract.

3. Name and address of test facility when testing is performed by the Government.

4. Shipping instructions when applicable.

5. Time required for the Government to notify the supplier whether or not to proceed with production.

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subject to some physical mixing operation intended to make the final product substantially uniform.

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6.3 Precaution. IMPORTANT. For procurement of this herbicide for use on lands owned by or otherwise managed as military installations, use Federal Specification O-H-200. Herbicides procured by MIL-H-51147 must not be diverted to domestic use.

Custodian:

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Preparing Activity:

Army - MU

Project Number 6840-A039

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